
Assignment 8 CS-2424 Bolatbekov Nurzhas

Part 1.

Monitoring of Hard Disk

Tasks: 1.

Follow instructions from videos Lecture 9 and Practice 9 from learn.astanait.edu.kz

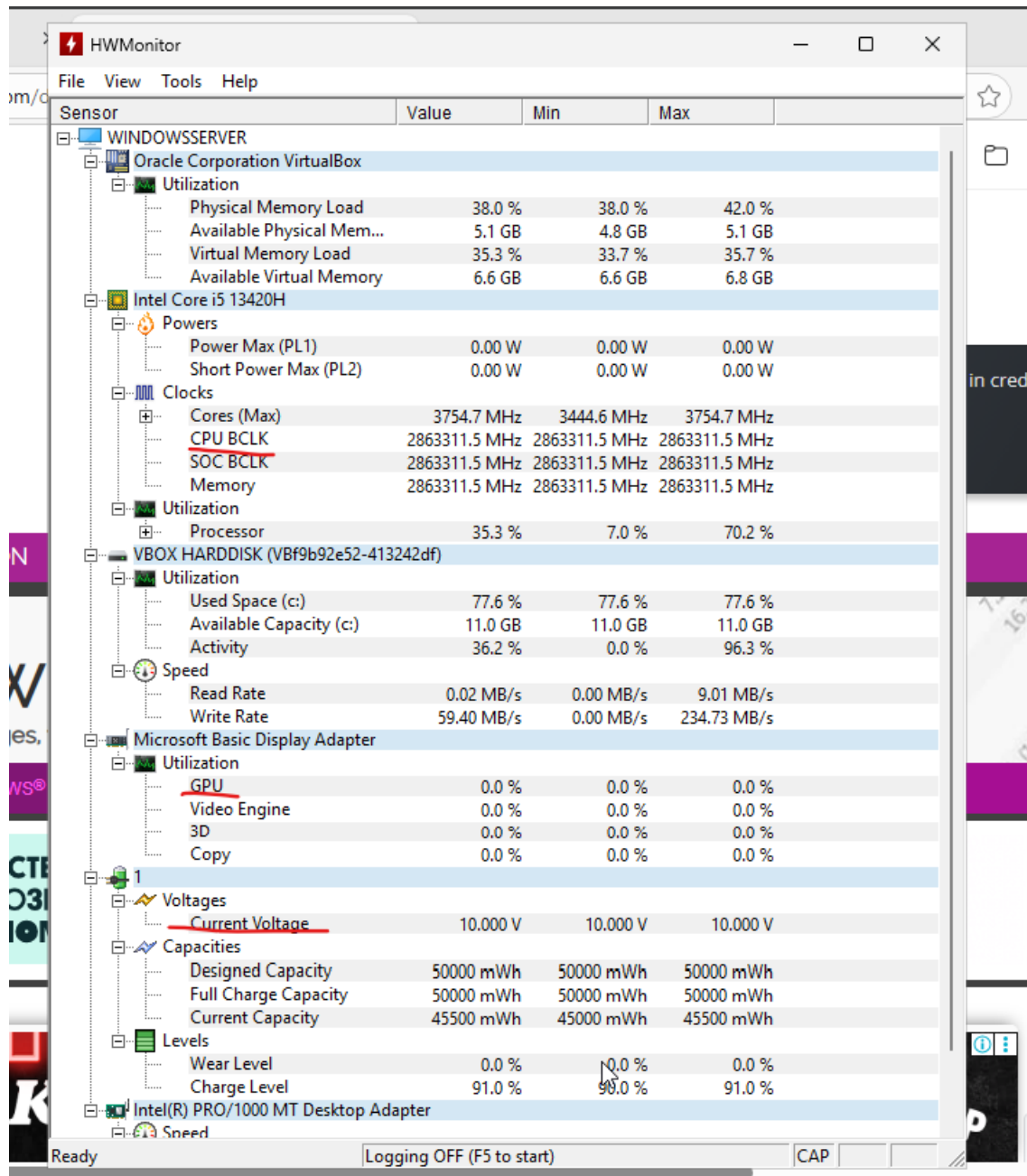
Step 1. Download and Install HWMonitor:

The screenshot displays the HWMonitor application window, which is a system monitoring tool. The interface is divided into several sections. On the left, there is a sidebar with the 'CPUID' logo and navigation tabs for 'SOFTWARE' and 'VALIDATION'. The main area shows a tree view of system sensors. The 'WINDOWSSERVER' sensor is expanded, showing sub-sensors for 'Oracle Corporation VirtualBox', 'Intel Core i5 13420H', and 'VBOX HARDDISK (VBF9b92e52-413242df)'. The 'VBOX HARDDISK' sensor is further expanded, showing 'Utilization' (Used Space: 77.6%, Available Capacity: 11.0 GB, Activity: 0.1%), 'Speed' (Read Rate: 0.00 MB/s, Write Rate: 0.10 MB/s), and 'Microsoft Basic Display Adapter'. The 'Intel Core i5 13420H' sensor is also expanded, showing 'Powers' (Power Max: 0.00 W, Short Power Max: 0.00 W), 'Clocks' (Cores: 3754.7 MHz, CPU BCLK: 2863311.5 MHz, SOC BCLK: 2863311.5 MHz, Memory: 2863311.5 MHz), and 'Utilization' (Processor: 12.4%). The 'Intel(R) PRO/1000 MT Desktop Adapter' sensor is also expanded, showing 'Utilization' (GPU: 0.0%, Video Engine: 0.0%, 3D: 0.0%, Copy: 0.0%), 'Voltages' (Current Voltage: 10.000 V), 'Capacities' (Designed Capacity: 50000 mWh, Full Charge Capacity: 50000 mWh, Current Capacity: 45000 mWh), and 'Levels' (Wear Level: 0.0%, Charge Level: 90.0%). The bottom status bar shows 'Ready' and 'Logging OFF (F5 to start)'.

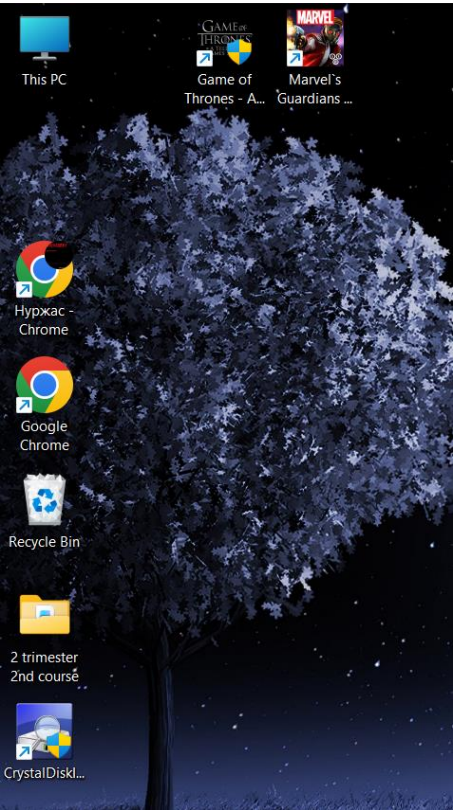
Sensor	Value	Min	Max
WINDOWSSERVER			
Oracle Corporation VirtualBox			
Utilization			
Physical Memory Load	41.0 %	41.0 %	42.0 %
Available Physical Mem...	4.9 GB	4.8 GB	4.9 GB
Virtual Memory Load	34.5 %	34.5 %	35.3 %
Available Virtual Memory	6.7 GB	6.6 GB	6.7 GB
Intel Core i5 13420H			
Powers			
Power Max (PL1)	0.00 W	0.00 W	0.00 W
Short Power Max (PL2)	0.00 W	0.00 W	0.00 W
Clocks			
Cores (Max)	3754.7 MHz	3444.6 MHz	3754.7 MHz
CPU BCLK	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
SOC BCLK	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
Memory	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
Utilization			
Processor	12.4 %	7.0 %	27.1 %
VBOX HARDDISK (VBF9b92e52-413242df)			
Utilization			
Used Space (c:)	77.6 %	77.6 %	77.6 %
Available Capacity (c:)	11.0 GB	11.0 GB	11.0 GB
Activity	0.1 %	0.0 %	3.9 %
Speed			
Read Rate	0.00 MB/s	0.00 MB/s	1.23 MB/s
Write Rate	0.10 MB/s	0.00 MB/s	0.26 MB/s
Microsoft Basic Display Adapter			
Utilization			
GPU	0.0 %	0.0 %	0.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	0.0 %	0.0 %	0.0 %
Copy	0.0 %	0.0 %	0.0 %
1			
Voltages			
Current Voltage	10.000 V	10.000 V	10.000 V
Capacities			
Designed Capacity	50000 mWh	50000 mWh	50000 mWh
Full Charge Capacity	50000 mWh	50000 mWh	50000 mWh
Current Capacity	45000 mWh	45000 mWh	45000 mWh
Levels			
Wear Level	0.0 %	0.0 %	0.0 %
Charge Level	90.0 %	90.0 %	90.0 %
Intel(R) PRO/1000 MT Desktop Adapter			
Speed			

Step 2. Monitor Hardware Using HWMonitor:

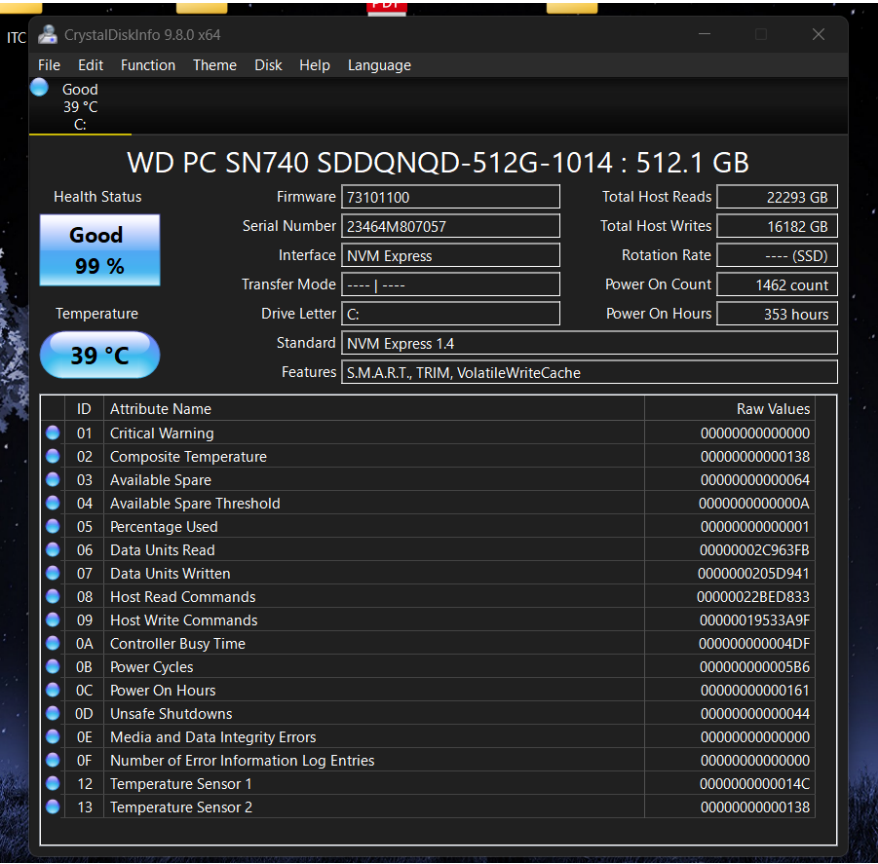
CPU (Processor) GPU (Graphics Card) RAM (Memory) - Record and analyze temperature, voltage and etc.



Step 3. Download and Install CrystalDiskInfo:



Step 4. Monitor Disk Health Using CrystalDiskInfo:



CrystalDiskInfo showed that the disk health status is 'Good' with no critical SMART warnings. Disk temperature remained within normal operating range

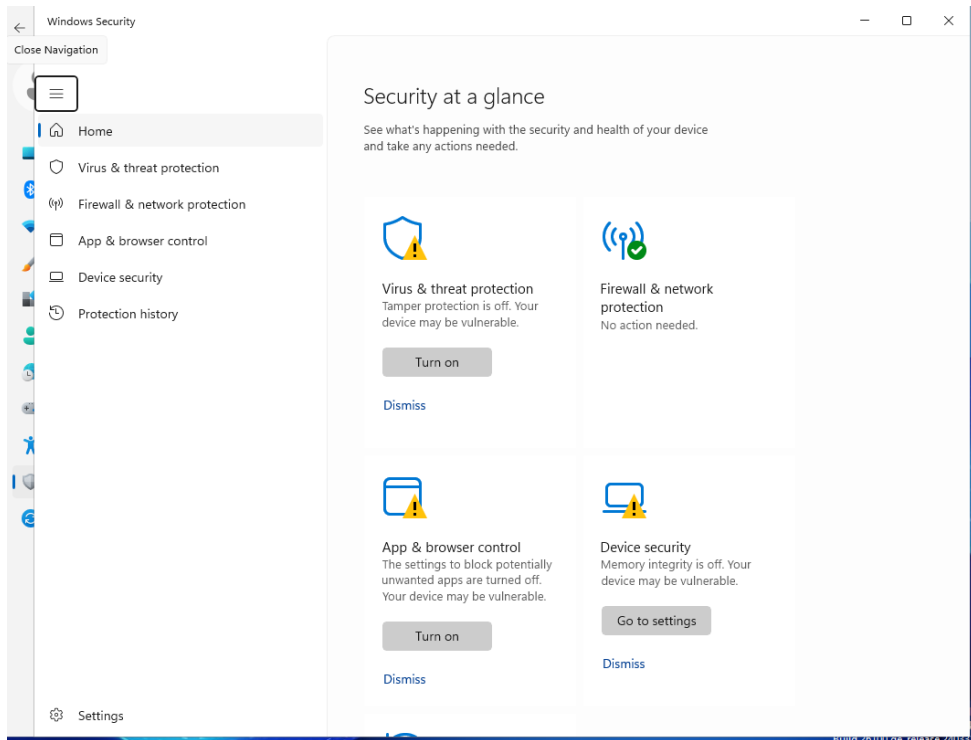
Part 2.

Setting up Windows Firewall with Advanced Security on Windows Server

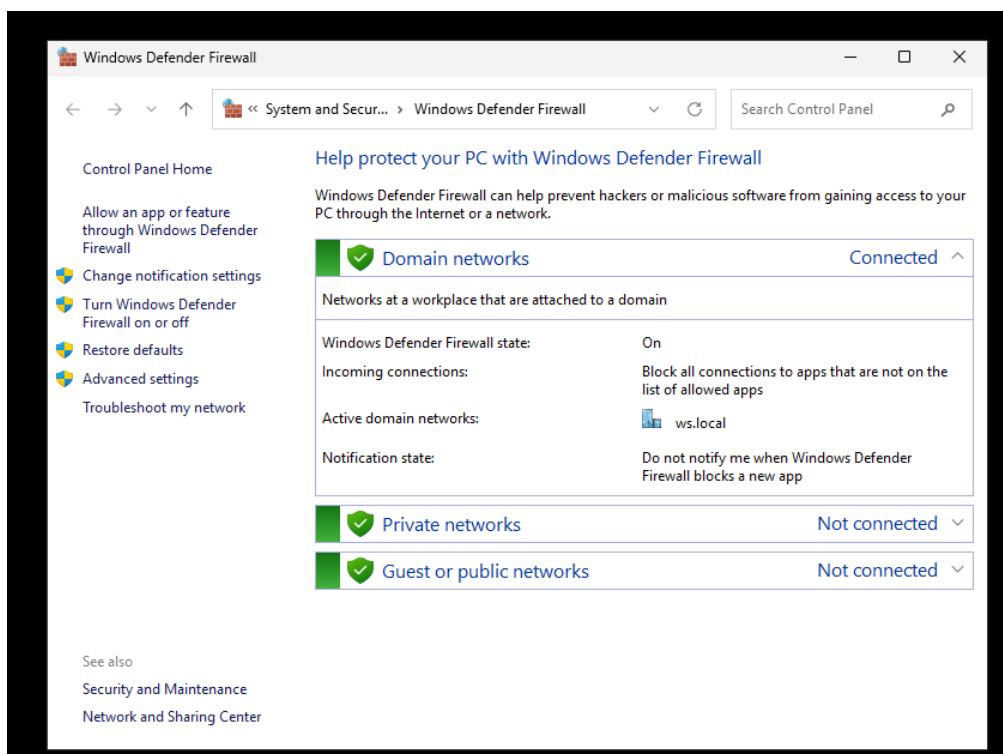
Tasks: 1.

Follow instructions from video Lecture and Practice 10 from learn.astanait.edu.kz

First, we go to Windows Security tab and observe there some features (we can turn the on or off as we wish):

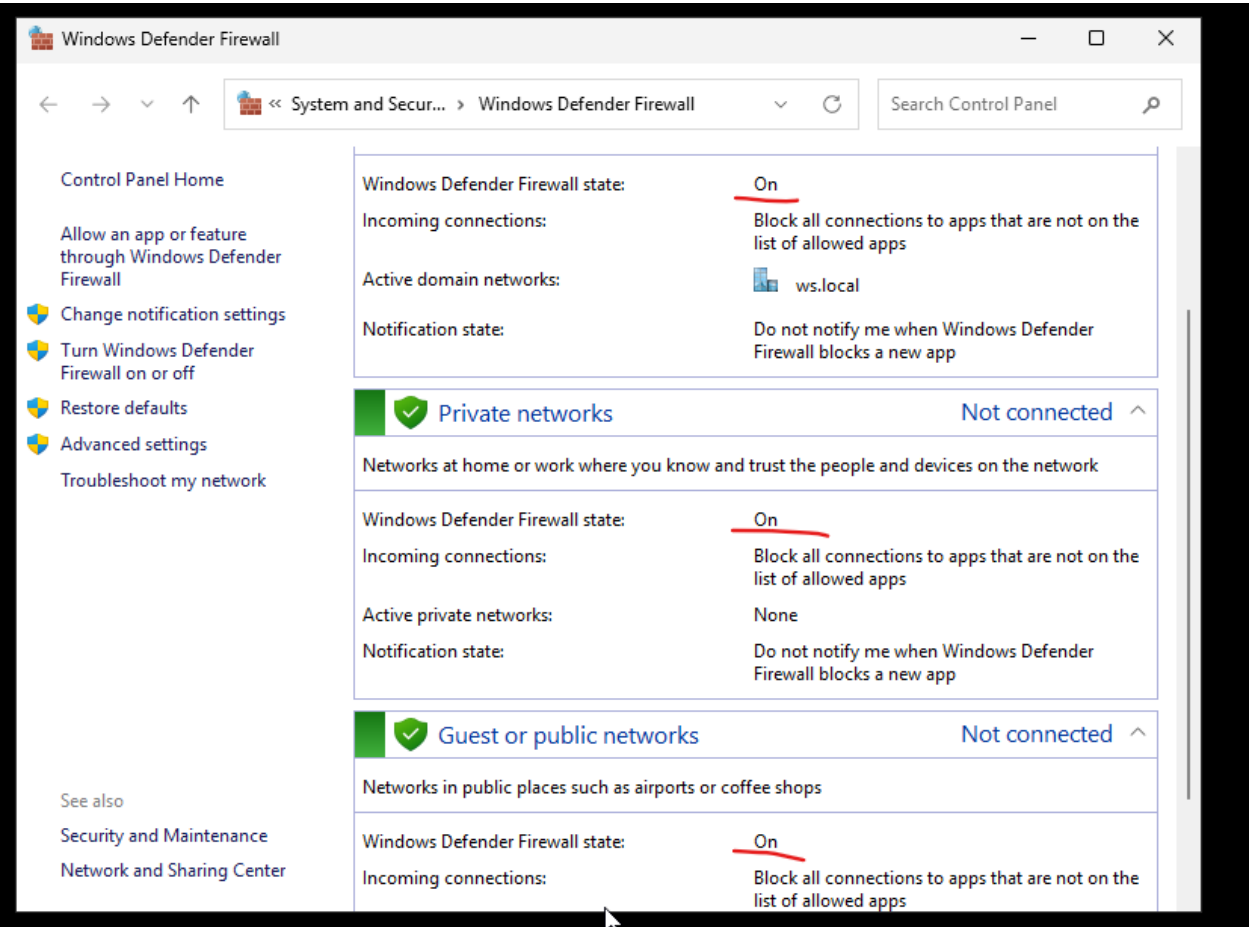


1. Open Firewall using the command `firewall.cpl` in the run dialog box (Win +R).

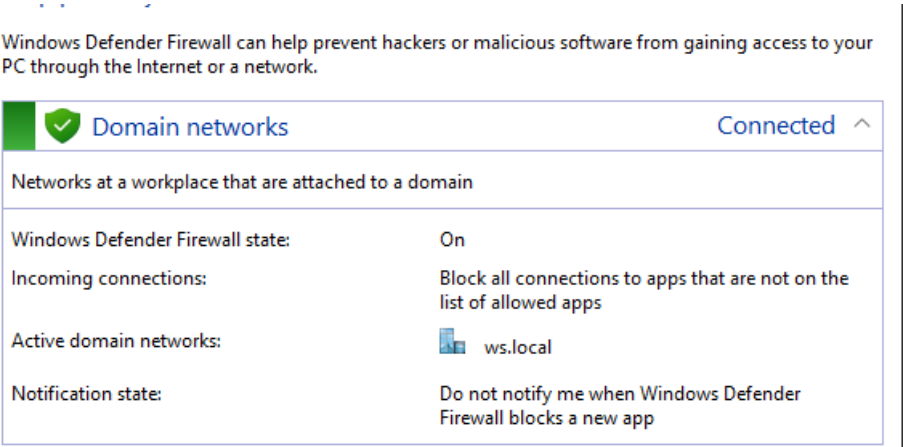


2.Under firewall settings, there are three sections called Domain, Private networks and Guest or public networks with green shields to the left of them, which says that the firewall is active. If they are on red shields, then the firewall is disabled.

Mine is active

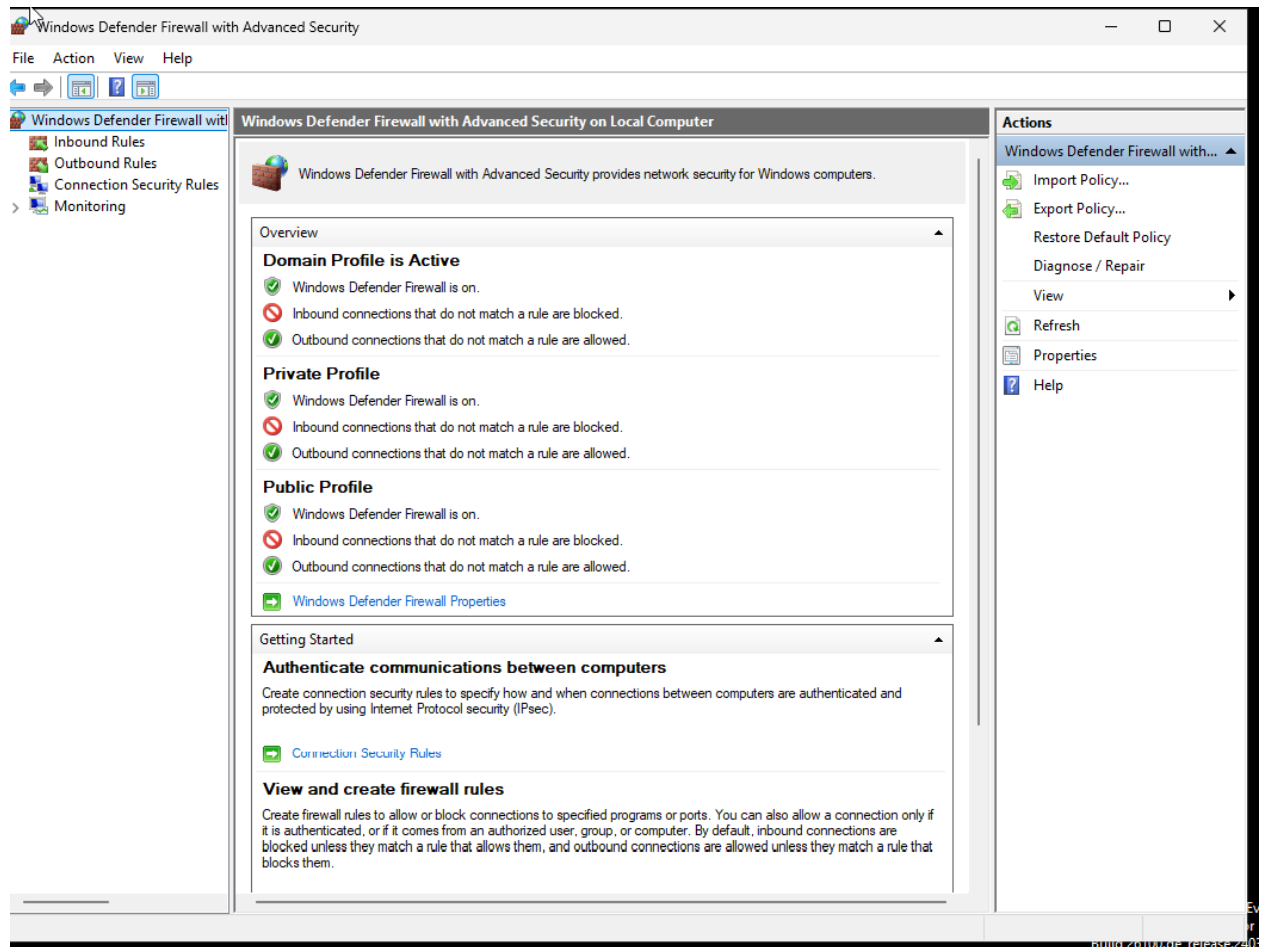


3.Click on either of these sections, it will prompt a drop-down menu with details about the current public or private networks.



4. Now select the Advanced settings option from the left-hand side of the page. The other way to open Advanced settings is to click on Tools in Server Manager and click on Windows Defender

Opened Windows Defender Firewall with Advanced Security to configure inbound and outbound traffic rules



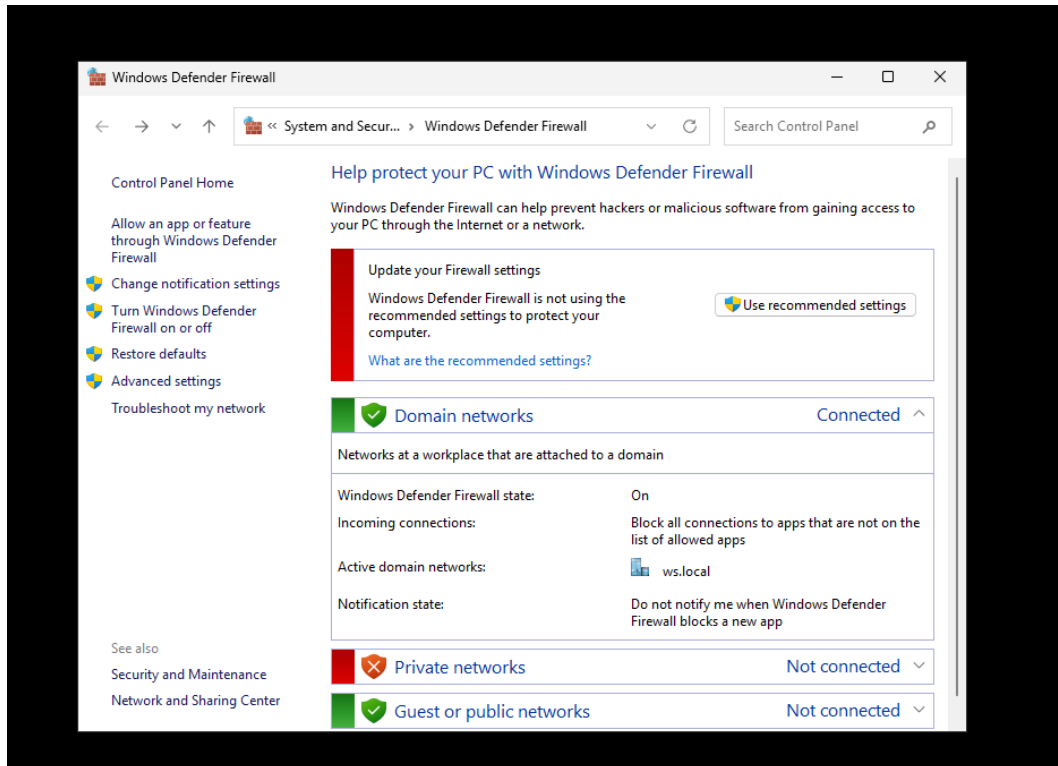
- *Inbound Rules*: Which incoming connections are automatically allowed.
- *Outbound Rules*: Which outgoing connections are automatically allowed.
- *Connection Security Rules*: Baselines for which connections your computer will allow and which ones it will block.
- *Monitoring*: This is an overview of your firewall's basic monitoring guidelines.

Part 3.

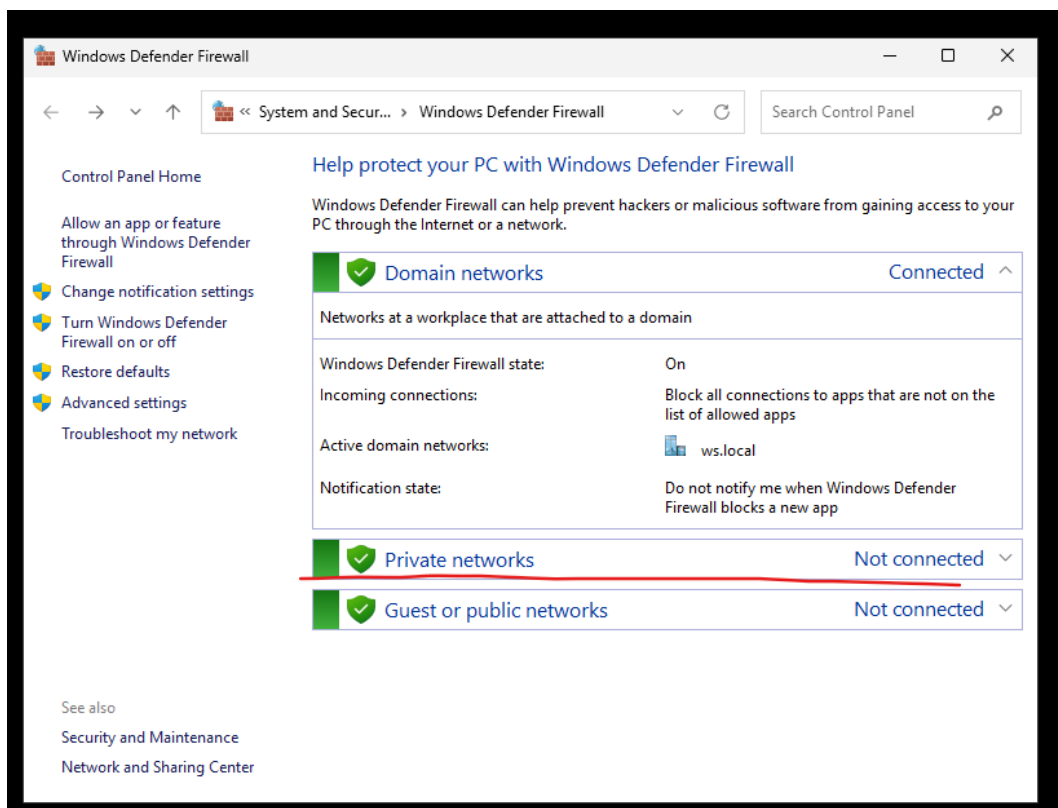
Tasks

1. Turn off and on Firewall for private profile

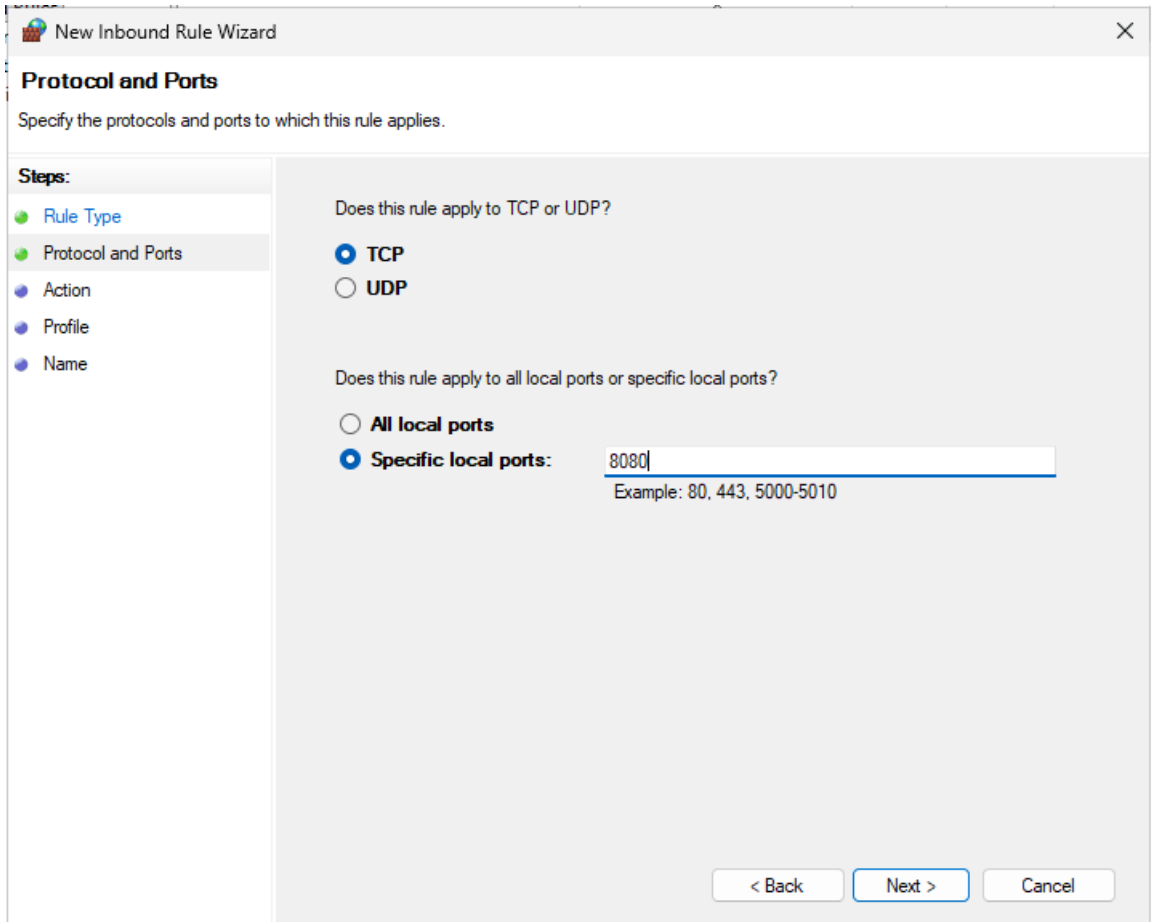
Private firewall profile was temporarily disabled for testing purposes.



Private firewall profile was enabled again to restore network protection.



2. Create an inbound rule that allows TCP traffic on port 8080



New Inbound Rule Wizard

Protocol and Ports

Specify the protocols and ports to which this rule applies.

Steps:

- Rule Type
- Protocol and Ports**
- Action
- Profile
- Name

Does this rule apply to TCP or UDP?

☒ TCP

☐ UDP

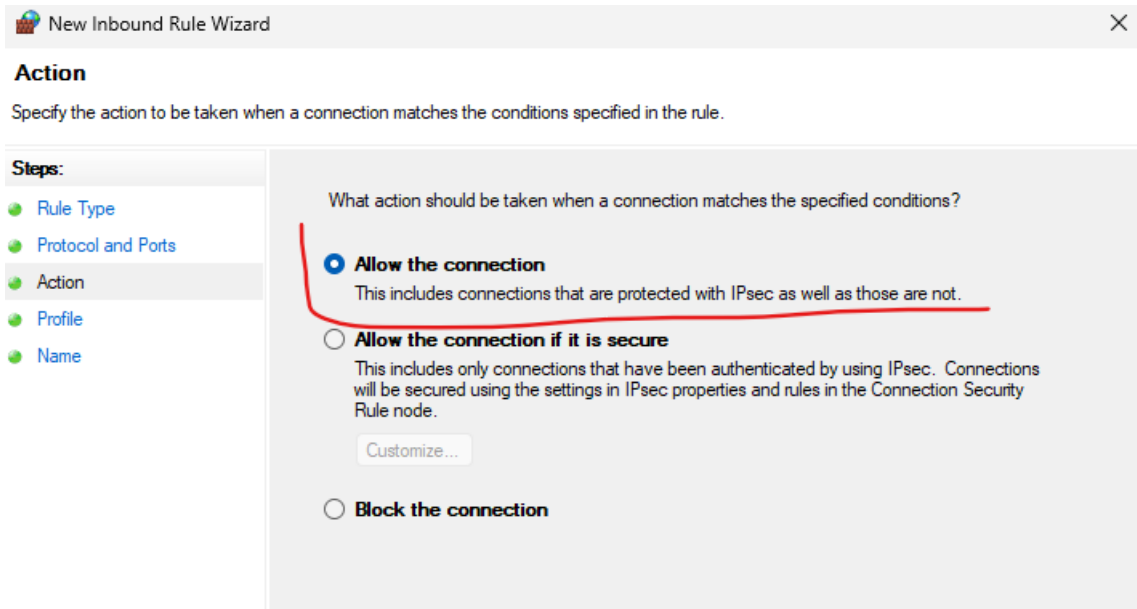
Does this rule apply to all local ports or specific local ports?

☐ All local ports

☒ Specific local ports:

Example: 80, 443, 5000-5010

< Back Next > Cancel



New Inbound Rule Wizard

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

- Rule Type
- Protocol and Ports
- Action**
- Profile
- Name

What action should be taken when a connection matches the specified conditions?

☒ **Allow the connection**

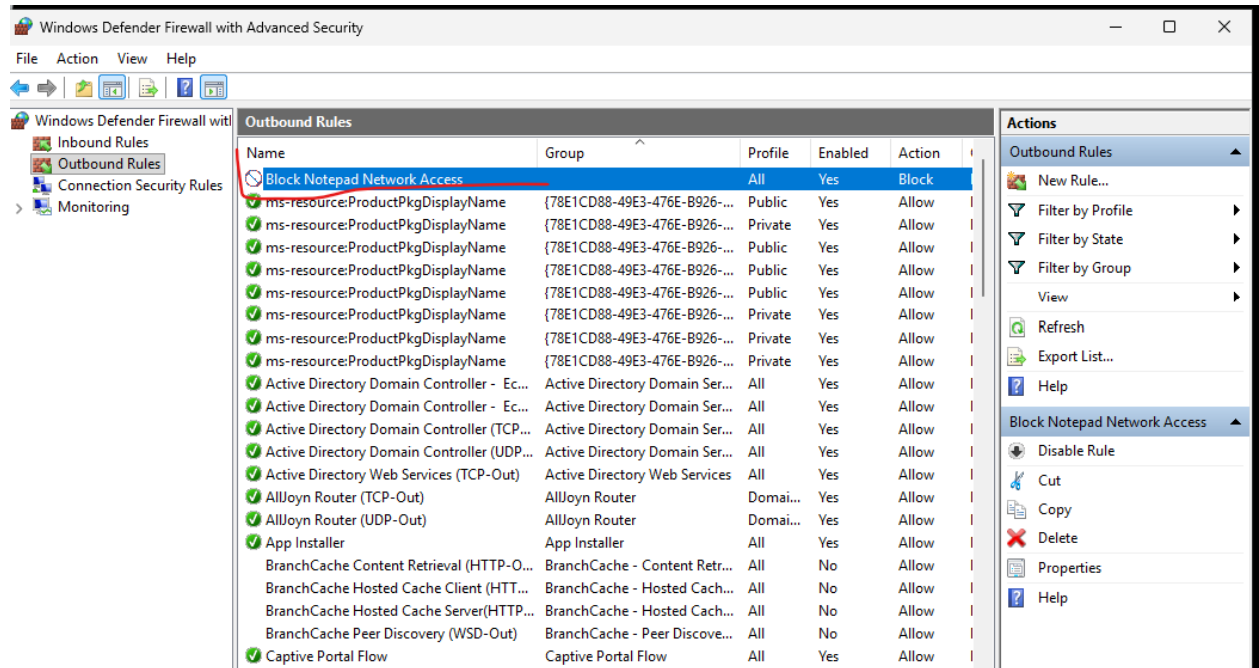
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**

This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

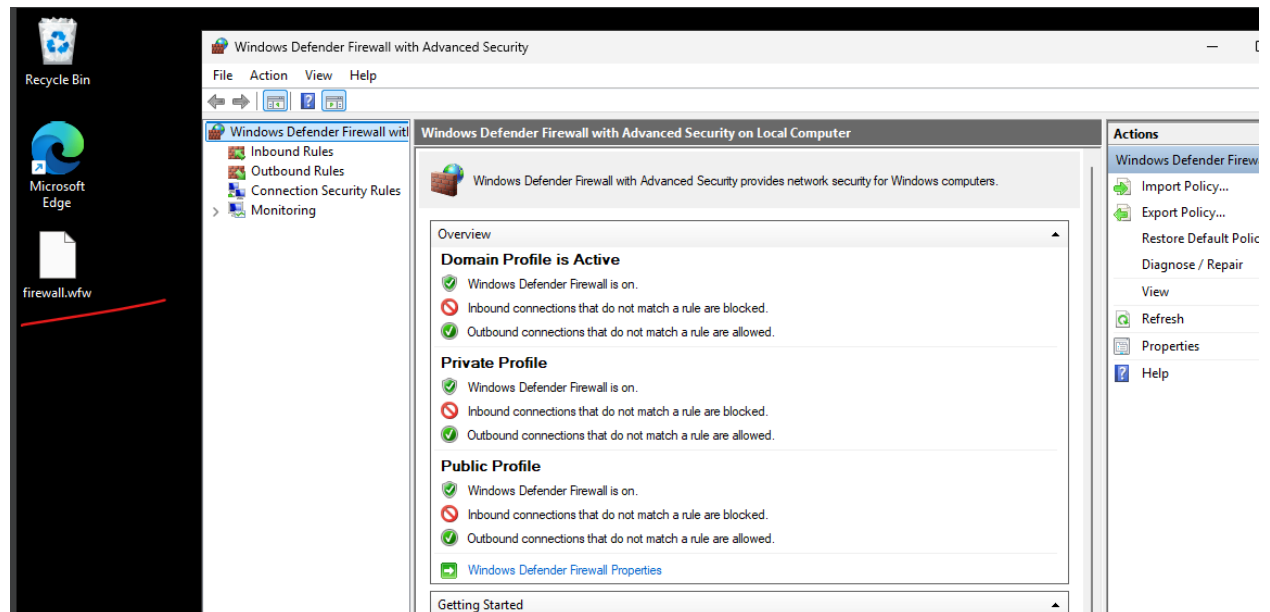
☐ **Block the connection**

Saved it:

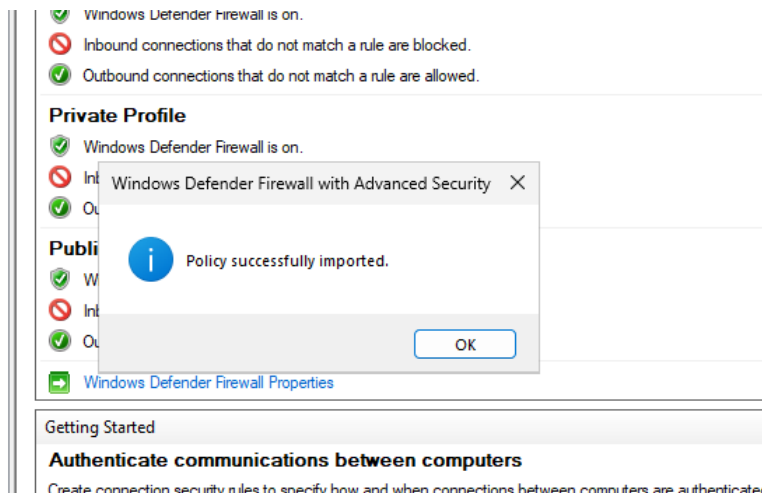
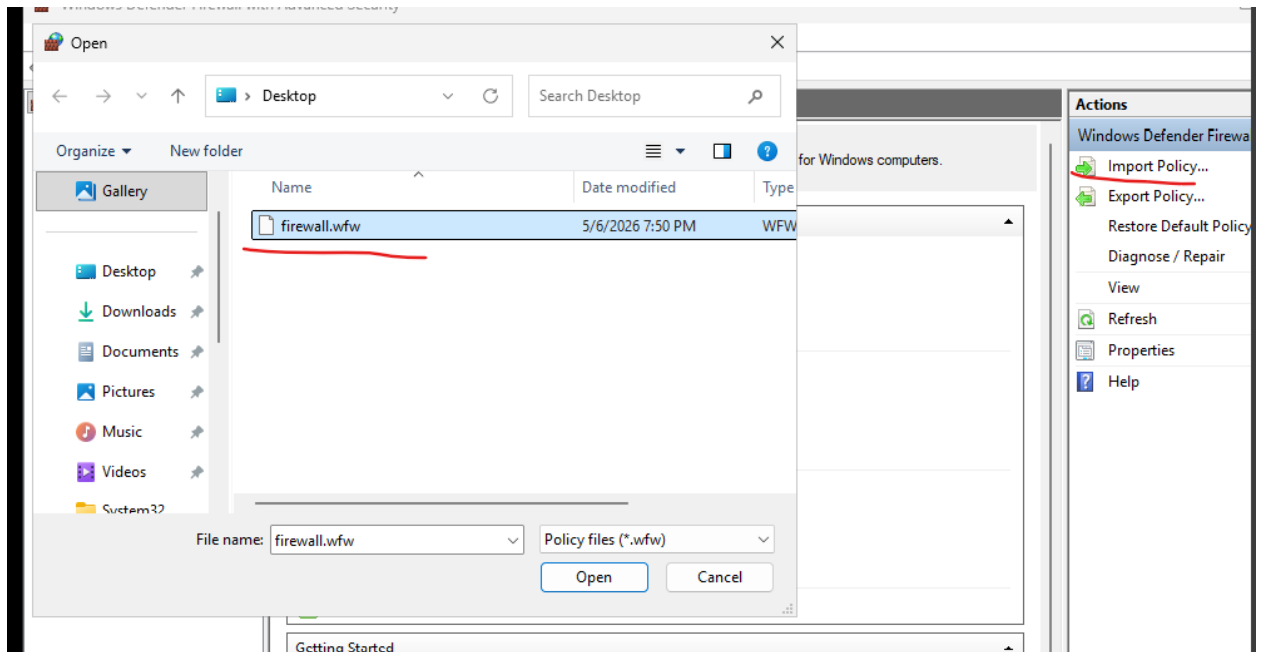


4. Export the current firewall policy to a file and then import a firewall policy from that file.

Exported Policy:



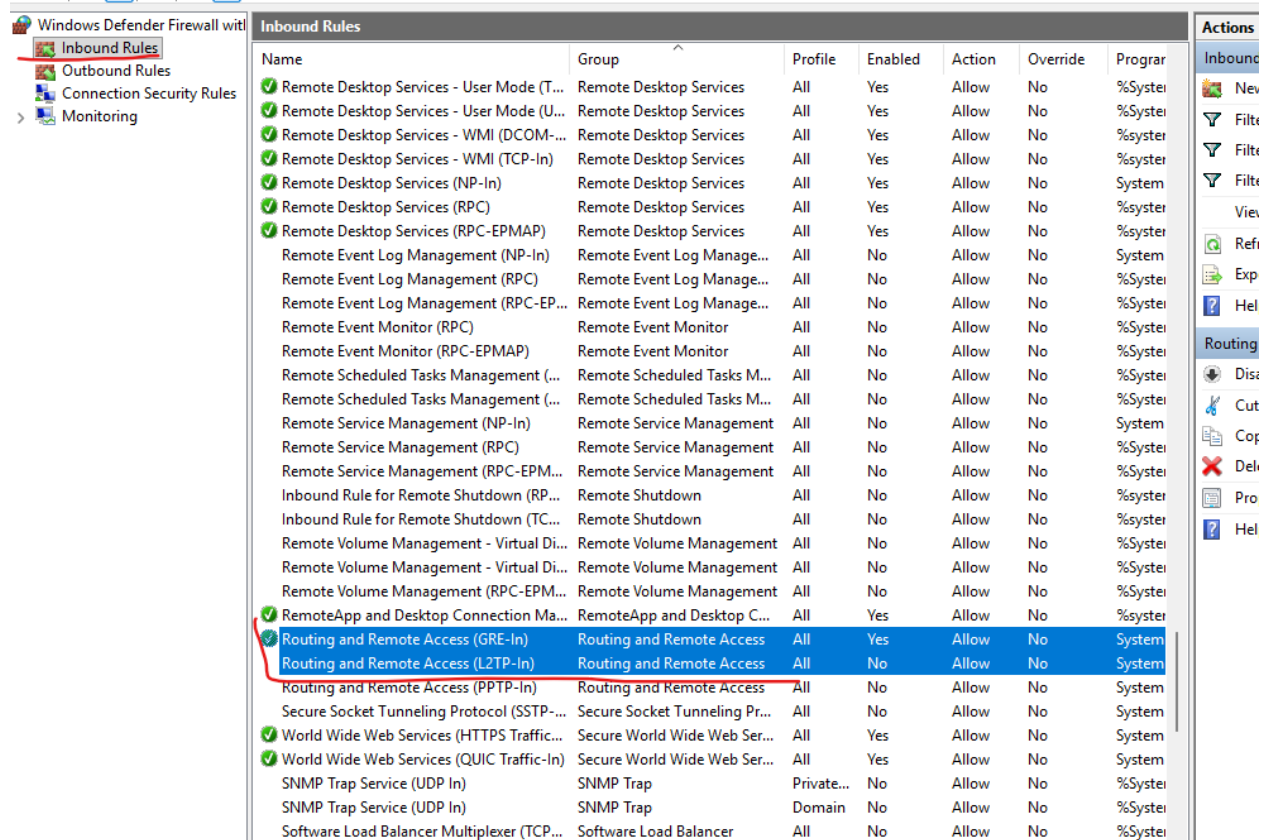
Import Policy:



Exported the current firewall configuration to a .wfw file and successfully imported it back into the system

5. Enable Routing and Remote Access rule

Enabled the Routing and Remote Access firewall rule to permit RRAS-related network traffic.



Windows Defender Firewall with Inbound Rules

Name	Group	Profile	Enabled	Action	Override	Program
Remote Desktop Services - User Mode (T...	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Desktop Services - User Mode (U...	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Desktop Services - WMI (DCOM-...	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Desktop Services - WMI (TCP-In)	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Desktop Services (NP-In)	Remote Desktop Services	All	Yes	Allow	No	System
Remote Desktop Services (RPC)	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Desktop Services (RPC-EPMAP)	Remote Desktop Services	All	Yes	Allow	No	%System...
Remote Event Log Management (NP-In)	Remote Event Log Manage...	All	No	Allow	No	System
Remote Event Log Management (RPC)	Remote Event Log Manage...	All	No	Allow	No	%System...
Remote Event Log Management (RPC-EP...	Remote Event Log Manage...	All	No	Allow	No	%System...
Remote Event Monitor (RPC)	Remote Event Monitor	All	No	Allow	No	%System...
Remote Event Monitor (RPC-EPMAP)	Remote Event Monitor	All	No	Allow	No	%System...
Remote Scheduled Tasks Management (...)	Remote Scheduled Tasks M...	All	No	Allow	No	%System...
Remote Scheduled Tasks Management (...)	Remote Scheduled Tasks M...	All	No	Allow	No	%System...
Remote Service Management (NP-In)	Remote Service Management	All	No	Allow	No	System
Remote Service Management (RPC)	Remote Service Management	All	No	Allow	No	%System...
Remote Service Management (RPC-EPM...	Remote Service Management	All	No	Allow	No	%System...
Inbound Rule for Remote Shutdown (RP...	Remote Shutdown	All	No	Allow	No	%System...
Inbound Rule for Remote Shutdown (TC...	Remote Shutdown	All	No	Allow	No	%System...
Remote Volume Management - Virtual Di...	Remote Volume Management	All	No	Allow	No	%System...
Remote Volume Management - Virtual Di...	Remote Volume Management	All	No	Allow	No	%System...
Remote Volume Management (RPC-EPM...	Remote Volume Management	All	No	Allow	No	%System...
RemoteApp and Desktop Connection Ma...	RemoteApp and Desktop C...	All	Yes	Allow	No	%System...
Routing and Remote Access (GRE-In)	Routing and Remote Access	All	Yes	Allow	No	System
Routing and Remote Access (L2TP-In)	Routing and Remote Access	All	No	Allow	No	System
Routing and Remote Access (PPTP-In)	Routing and Remote Access	All	No	Allow	No	System
Secure Socket Tunneling Protocol (SSTP-...	Secure Socket Tunneling Pr...	All	No	Allow	No	System
World Wide Web Services (HTTPS Traffic...	Secure World Wide Web Ser...	All	Yes	Allow	No	System
World Wide Web Services (QUIC Traffic-In)	Secure World Wide Web Ser...	All	Yes	Allow	No	System
SNMP Trap Service (UDP In)	SNMP Trap	Private...	No	Allow	No	%System...
SNMP Trap Service (UDP In)	SNMP Trap	Domain	No	Allow	No	%System...
Software Load Balancer Multiplexer (TCP...	Software Load Balancer	All	No	Allow	No	%System...

THE END

